

SSH Honeypot Project – Attack Simulation and Monitoring

Introduction

A honeypot is a decoy system designed to attract attackers and record their activities. In this project, a simple SSH honeypot was built using Python to simulate a vulnerable SSH service. The honeypot listens for incoming connections and logs attacker behavior such as connection attempts and SSH handshake data. This helps researchers understand how attackers scan networks and attempt to gain access.

Objective

The objective of this project was to create a basic honeypot environment capable of detecting malicious traffic. The honeypot was deployed on an Ubuntu machine, while Kali Linux was used to simulate an attacker. All incoming connection attempts, IP addresses, and interaction data are recorded in logs for analysis.

Technologies Used

- Python (socket programming)
- Ubuntu Linux (honeypot host)
- Kali Linux (attack simulation)
- Nmap (network scanning)
- Hydra (SSH brute-force testing)

System Architecture

The lab environment consists of two virtual machines connected through a local virtual network. The Ubuntu machine runs the honeypot program listening on port 2222, while the Kali machine performs attack simulations. All traffic reaching the honeypot is logged and analyzed to understand attacker behavior.

Implementation

1. A Python socket server was created to listen on port 2222.
2. The program records incoming connections and attacker IP addresses.
3. A fake SSH banner is sent to simulate a real SSH server.
4. Attack simulations were conducted using SSH, Nmap scans, and Hydra brute-force attempts.
5. Logs are stored locally for monitoring and analysis.

Attack Simulation

The attacker machine (Kali Linux) performed several simulated attacks including direct SSH login attempts, port scanning using Nmap, and brute-force password attacks using Hydra with the RockYou wordlist. These actions mimic real-world automated attacks commonly seen on exposed servers.

Results

The honeypot successfully captured connection attempts and recorded attacker IP addresses along with SSH handshake information. Even though the honeypot does not provide full SSH authentication, it effectively logs reconnaissance and brute-force attempts.

Conclusion

This project demonstrates how honeypots can be used to monitor malicious network activity in a controlled lab environment. By simulating a vulnerable SSH service, the honeypot provides insight into attacker techniques and helps security researchers study real-world attack patterns.